

Machine-learning based detection of high-frequency gravitational waves on a localized VAE latent representation

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High-frequency gravitational waves (HFGWs) from mergers of sub-solar-mass primordial black holes (PBHs) can couple to microwave-cavity experiments such as SUPAX. The detection of such signals can be difficult as the induced signals are deeply buried in detector noise. I develop a variational-autoencoder (VAE) with a classifier directly operating on the latent space, that detects such signals on a single, explicit, low-dimensional latent code, shaped simultaneously by a generative decoder and by specifically designed loss terms. I compare two variants differing only in whether the latent-classifier additionally sees the output of the decoder. Detection is quantified by the peak SNR required for 95% event-detection efficiency across false-positive rates and PBH masses. The latent-only variant performs best overall, and the hardest mass to detect, $10^{-10} M_{\odot}$, coincides with a cavity time-scale matching $\Delta t = \tau$ that also seems to be imprinted in the latent representation. Throughout, I highlight open directions for improving the architecture.

I. BACKGROUND AND PRIOR WORK

The detection of gravitational waves (GWs) is experimentally challenging, particularly in the high-frequency regime, where the expected signal strains are extremely small and conventional interferometric techniques rapidly lose sensitivity [1]. Microwave-cavity experiments, like the SUPAX experiment [2], offer a promising alternative for probing high-frequency gravitational waves (HFGWs) through their coupling to electromagnetic cavity modes via the inverse Gertsenshtein effect (see [3] for a derivation); the resulting signals are, however, typically deeply buried in noise and subject to instrumental and environmental effects. As in the preceding report [4], that probed the usage of a Convolutional Neural Net (CNN) combined with a Long Short-Term Memory [5] (LSTM) model, operated as a classifier [4], this work investigates machine-learning-based methods to identify HFGW-induced signatures in microwave-cavity data from the SUPAX experiment [2], where the HFGWs considered stem from mergers of sub-solar-mass primordial black holes (PBHs) [6].

The preceding report [4] established a hybrid convolutional-recurrent (CNN-LSTM) classifier that performs binary signal/noise classification and achieves fewer than one false-positive per year while generalizing in detection across several orders of magnitude in PBH mass [7]. That model treats the learned internal representation purely as an intermediate quantity. The present work instead places the internal representation at the centre of the architecture: a variational autoencoder (VAE) [8] is used so that classification is performed on a single, explicit, low-dimensional latent representation, which is simultaneously shaped by a generative decoder and by physics-motivated loss terms. The aim is a representation that is not only discriminative but also compressed, inspectable, and ultimately usable for future applications such as HFGW signal parameter estimation. Two variants of this design are considered and compared,

differing in whether the decoder reconstruction is also exposed to the classifier (see Sec. II) or not.

The remainder of this report motivates the architectural choice and introduces the two model variants compared here (see Sec. II), describes the revised signal simulation compared to the preceding report (Sec. III), explains the training strategy and loss design (Sec. IV), as well as the evaluation methodology (Sec. V), and concludes with a results section (Secs. VI–VII). Finally, the report is closed with a first parameter-estimation prototype (Sec. VIII), a Summary IX, a Conclusion X and an Outlook XI.

Throughout the report I highlight open directions for how the VAE latent-classifier architecture can be further analysed and potentially improved.

II. CHOOSING A VAE LATENT CLASSIFIER ARCHITECTURE

A schematic of the chosen architectures is shown in Fig. 1 and motivated below.

My primary motivation for adopting a VAE-based architecture is its latent space: In a general neural network (like an LSTM or a Transformer [9]), the internal representation evolves gradually from layer to layer, and there is no single, architecturally defined location at which the full, compressed representation of the input resides. A VAE, by contrast, provides exactly such a location: the latent code (= a compressed abstract representation of a given input) is a designated point in the architecture through which all of the retained information must pass. One can, of course, inspect the intermediate representations of any deep network by probing individual layers and their weights, so the ability to look inside is not unique to the VAE. What is unique is that the VAE gives a *localised* representation - a single, well-defined place where I know that all of the learned compressed

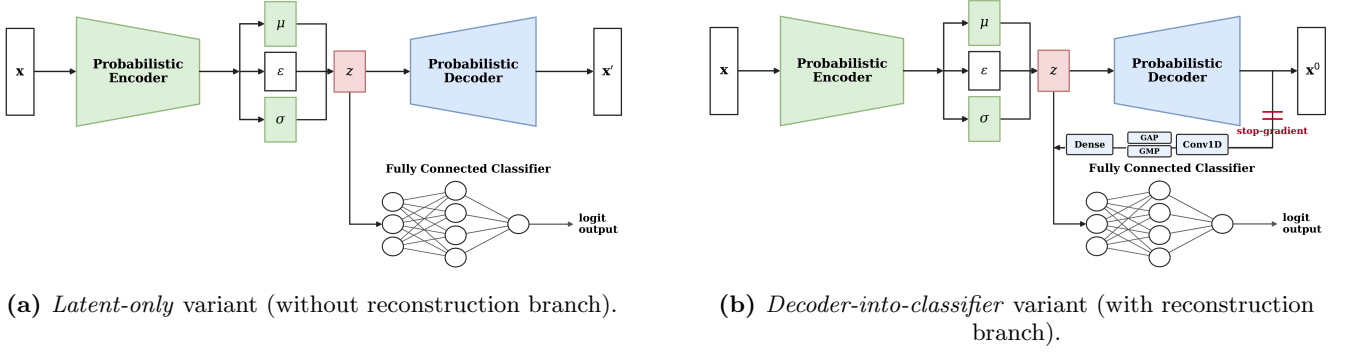


FIG. 1: Schematic of the two VAE-classifier architectures compared in this work. In both variants an input window $\mathbf{x}$ is mapped by the *probabilistic encoder* to the parameters of an approximate multi-dimensional Gaussian posterior with mean $\mu = z_{\text{mean}}$ and standard deviation $\sigma = z_{\text{log var}}$. A latent code is then formed by the reparameterisation trick, $z = \mu + \sigma \odot \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \mathbb{I})$ is an auxiliary noise vector; this keeps the sampling differentiable so that gradients can flow back through the encoder (see Sec. 2.4 of [8] for further details). The latent code $z \in \mathbb{R}^d$, with d the latent dimension, is passed to the *probabilistic decoder*, which produces the reconstruction $\mathbf{x}'$ (in this work the injected clean signal), and is also fed to a *fully-connected classifier* whose hidden layers non-linearly combine the latent features and whose single output neuron yields the raw classification logit. **(a)** In the *latent-only* variant the classifier consumes the latent code z alone, so the decoder and classifier interact solely through the shared representation z . **(b)** In the *decoder-into-classifier* variant the classifier additionally receives features derived from the reconstruction $\mathbf{x}'$: a small Conv1D branch processes the reconstruction, its feature maps are reduced by global average pooling (GAP) and global max pooling (GMP) into a position-invariant summary, and a Dense layer turns this into a feature vector that is concatenated with the latent features at the classifier input. This “reconstruction branch” is detached by a *stop-gradient* (double red bar), so the reconstruction informs the classification decision while the classifier’s gradients do not flow back into the decoder; the decoder is therefore shaped only by the reconstruction, KL and correlation terms.

information resides - rather than a representation spread across many layers, where it is not clear where one should even look.

In addition, the compression matters as input data can often be redundant (as it can be for the case of a signal buried in noise): distilling it into one compact, canonical code is therefore desirable. A VAE addresses this directly: its low-dimensional latent code acts as an information bottleneck [10] that is forced to discard redundancy and retain a compact, meaningful representation of the *measured signal* itself.

It should, however, be stressed that compression alone does not guarantee that the retained latent code aligns with physically meaningful factors; that alignment is rather imposed by the supervised objectives described below.

a. A discriminative latent. Because the classifier reads the latent code, the classification loss back-propagates into the encoder and can shape the latent so that noise-only and signal-plus-noise windows are mapped to different regions of the latent space. This is potentially advantageous in two directions at once: the classifier gradient can make the latent more expressive and better organised, while the classifier itself benefits from operating on an organized, compressed representation rather than on the raw window. Beyond the benefit for the classifier, the classifier gradients have empirically

been found to aid the decoder in reconstructing the signal, as examined by the gradient-based ablation study explained in Sec. IV A .

b. A physics-shapeable latent. Finally, because the latent is a single explicit tensor, it can be shaped by additional loss terms. Reconstructing the injected *clean* signal (rather than the noisy input) forces the code to encode the signal structure itself.

Additionally, correlation and physics-motivated terms can be added; and regression heads can be attached that predict physical parameters from the same code. This directly opens a path towards physics-informed latents and parameter estimation, which may in turn aid classification due to more expressive latent codes.

The validity of this picture is bounded by the usual limitation of learned representations: the model learns a parametrisation of, and an approximation to, the generative process consistent with the training data.

c. A reconstruction-informed classifier. The description so far corresponds to the baseline model of Fig. 1a, in which the classifier reads the latent code z alone. I additionally consider the second variant of Fig. 1b, in which the decoder reconstruction $\mathbf{x}'$ is also made available to the classifier. In principle, this adds no new information: $\mathbf{x}'$ is a deterministic function of the *same* latent code z , so both classifier inputs ulti-

mately originate from the same bottleneck. The motivation is rather that the reconstruction offers a *different, non-linearly transformed view* of the same code - one expressed back in the signal (I/Q) domain instead of in the abstract latent space. Empirically, I found that the decoder reconstructs the injected signal well visually even at low SNRs, so the idea is to aid the classifier with this complementary, signal-aligned perspective of the same data. Concretely, the reconstruction is first detached by a stop-gradient and fed to a small convolutional branch operating on I , Q and the magnitude $|\mathbf{x}'| = \sqrt{I^2 + Q^2}$. Its final feature maps are reduced by global average pooling (GAP) and global max pooling (GMP), whose concatenation forms a position-invariant summary (a signal anywhere in the window contributes) that a dense layer turns into a feature vector. This vector is concatenated with the latent features before the fully-connected classifier (Fig. 1b). The stop-gradient ensures the classifier gradients shape only this auxiliary branch and never reach the decoder, to not interfere with the reconstruction task of the decoder itself¹. The two variants are compared in Sec. VI to isolate the effect of exposing the classifier to the decoded reconstruction.

III. SIGNAL SIMULATION

The generation of the synthetic HFGW signals follows the procedure of the preceding report [4] and thesis [7], to which the reader is referred for the full derivation. I only describe the changes relevant to this work here.

As before, each waveform is obtained as a frequency-domain chirp, whose chirp rate is set by the PBH mass, multiplied by the cavity Breit-Wigner (BW) response. Afterwards, the product of the BW function and chirp spectrum is transformed to the time domain via an inverse fast Fourier transform (iFFT), yielding a complex I/Q waveform that is injected into real detector noise at a prescribed peak SNR.

The principal change introduced in this work concerns the frequency grid on which the waveform is generated. Previously, a frequency span of 2 MHz was used; here, the frequency span was matched to the detector sampling rate of 14 MHz, so that the simulated waveform shares the detector time base. This modification increases the effective signal length by a factor of 7 in time domain and, as a consequence, changes the task of a local spike-like detection that spans less than $\frac{1}{10}$ of a window's length (provided the window consists of 1024 samples) to signals spanning more than $\frac{1}{2}$ (or more depending on the PBH mass) of a window's length.

During preprocessing, the exact injected signal component is additionally stored as a clean-signal target for

later use as a reconstruction target, further discussed in Sec. IV B.

A. Implications of the revised waveform sampling

In the following, I want to be explicit about the consequence of the change in frequency span and the impact it has for how the signal is detected, in combination with the relationship to the preceding study [4].

Matching the waveform frequency span to the detector sampling rate does not alter the *morphology* of the simulated signal: the underlying waveform is the same chirp-cavity response as before, now represented over roughly seven times as many samples, since its temporal support is longer at the fixed 14 MHz sampling rate. In other words, the change is one of temporal resolution and extent, not of shape.

In earlier studies [4] (*frequency span is not matched to the sampling rate*), the signal was compact enough to fit within the kernel of a single first-layer convolutional filter, so that detection was effectively matched-filter-like: a first-layer kernel could act as a learned template for the entire waveform, and detection reduced to finding where that template correlated strongly with the input. With the longer signal this is no longer possible without drastically increasing the kernel size (*which is a possible direction to try in the future!*).

Detection instead relies on the hierarchical structure of the convolutional stack, in which successive layers progressively enlarge the effective receptive field, so that only the deeper layers integrate over the full temporal extent of the signal.

I emphasise that this does not invalidate the results of the preceding report [4]. That study answered the same underlying question - whether a window contains a signal or only noise - but for a physically different family of signals, and it is therefore expected to generally yield a different answer. Its conclusions remain valid for the signals it considered; here I simply repose the question for the detector time scale matched longer waveform.

I further find that the longer, more realistic signal in fact *decreases* the needed SNR for 95% detection efficiency, shown in Sec. VI. A direct consequence of 7-times more samples is: more of a window is occupied by signal-bearing samples, which increases the evidence available to the network per window. One might suspect that this is purely an effect of the signal-to-window occupancy ratio, i.e. of how large a fraction of the window the signal fills. I do not find this explanation supported by previous studies: shortening the analysis window for the previously seven-times shorter signal, which would raise the same occupancy ratio, did not produce a comparable improvement in the legacy studies.

A possible explanation for the decrease in needed SNR for 95% detection efficiency is the following: Because the length of a signal in samples is set jointly by the sampling rate and by the duration of the cavity oscillations, a

¹ I did not try the variant without the stop-gradient to verify that this is indeed needed.

mismatch between the simulation and detector frequency grids samples the injected signal and the real noise on incompatible grids, superimposing physically inconsistent signal-noise combinations that can artificially inflate the task’s difficulty. Matching the two grids removes this inconsistency, so I consider it plausible that the reduced SNR required for detection arises from the grid alignment itself rather than from the relative occupancy of the window alone.

IV. NETWORK ARCHITECTURE, LOSS DESIGN AND TRAINING

This section presents the model architecture together with the training strategy and loss design, the latter two being integral parts of that architecture. As introduced in Sec. II, two variants of the architecture are considered: the *latent-only* model, in which the classifier reads the latent code alone (Fig. 1a), and the *decoder-into-classifier* model, in which the detached decoder reconstruction is additionally summarised by a small convolutional branch ($\text{Conv1D} \rightarrow \text{GAP/GMP} \rightarrow \text{Dense}$) and concatenated with the latent features at the classifier input (Fig. 1b). The two variants share the same encoder, decoder and classifier hyperparameters (HPs) and differ only in the presence of this reconstruction branch. Additionally, they are trained with an identical data pipeline and loss configuration, so that any difference in detection performance can be attributed to the branch alone. I note that this is generally not true, which is further discussed in Sec. IV C

The training data are prepared in the following order: First, signals are injected into the raw detector time-series as described in Sec. III, using a *no-overlap* injection mode in which the injected signals are separated by a margin large enough that no subsequent window can contain samples from two different signals; *each window therefore holds at most one injected event*. The resulting stream is then windowed into segments of 1024 samples. Finally, each window is z -score normalised and the windows are partitioned into training, validation and test sets. I use the no-overlap mode to guarantee clean per-window labels and to avoid ambiguous windows in which two signals could overlap or interfere. The exact hyperparameters used for training and fine-tuning are listed in Appendix A.

I train the model directly on the raw I and Q quadratures, rather than on the amplitude time-series used in the previous work [4]. The main motivation for this is that the I and Q quadratures retain the phase information, which increases the number of degrees of freedom available to the model, potentially aiding it. For future works considering lighter masses (starting at roughly $1 \times 10^{-11} M_\odot$) whose induced signal in the cavity is oscillatory, the I and Q quadrature make it possible to directly access the oscillatory parts of the induced signal. Additionally, signal reconstruction of the oscilla-

tions themselves can be performed via the decoder², by choosing the correlation + clean-MSE loss discussed in Sec. IV B.

With the data representation and the two architectural variants fixed, the remainder of this section turns to the training itself and the refinements it required. I begin with a gradient-based ablation study that probes how the classifier and the decoder interact through the shared latent code - the central design question underlying both variants - and then describe the loss design in Sec. IV B. Furthermore, the training and fine-tuning procedure itself is outlined in Sec. IV C.

A. Gradient-based ablation of the latent-classifier coupling

A central design question is whether the classifier (beyond just classification), acting on the latent space, contributes anything beyond what a plain VAE decoder already learns and, if so, whether its gradient cooperates with, or opposes, the reconstruction objective on the shared encoder.

To answer this, an ablation study was carried out to examine the impact of the classifier on the encoder. In every mode the encoder and the decoder pathway are held fixed: the decoder always receives the sampled latent

$$z = \mu + \sigma \odot \epsilon_{\text{dec}} \in \mathbb{R}^d, \quad d = \text{latent dimension}, \quad (1)$$

so that any difference in behaviour must originate from how the classifier loss reshapes the encoder. After $z_{\text{mean}}, z_{\text{log var}} = \text{encoder}(x)$, the classifier is fed one of the following:

- **concat_mean_logvar**: the deterministic concatenation $[z_{\text{mean}}, z_{\text{log var}}]$. The classifier never sees the reparameterisation noise; encoder gradients flow only through the z_{mean} and $z_{\text{log var}}$.
- **z_mean**: z_{mean} alone, an “even more deterministic” control that additionally starves the head of the σ information.
- **shared_sample**: the same sampled z that the decoder receives, so that the classifier loss propagates into the encoder through the reparameterisation with the *same* noise realisation ϵ_{dec} .
- **independent_sample**: $z_{\text{cls}} = \mu + \sigma \odot \epsilon_{\text{cls}}$, with a *freshly drawn* $\epsilon_{\text{cls}} \sim \mathcal{N}(0, \mathbb{I})$ independent of ϵ_{dec} . This isolates whether just the noise injection matters or whether the shared realisation specifically matters.

² This was explicitly found during training on lighter masses, which, however, was not done in detail and is therefore not presented in this work.

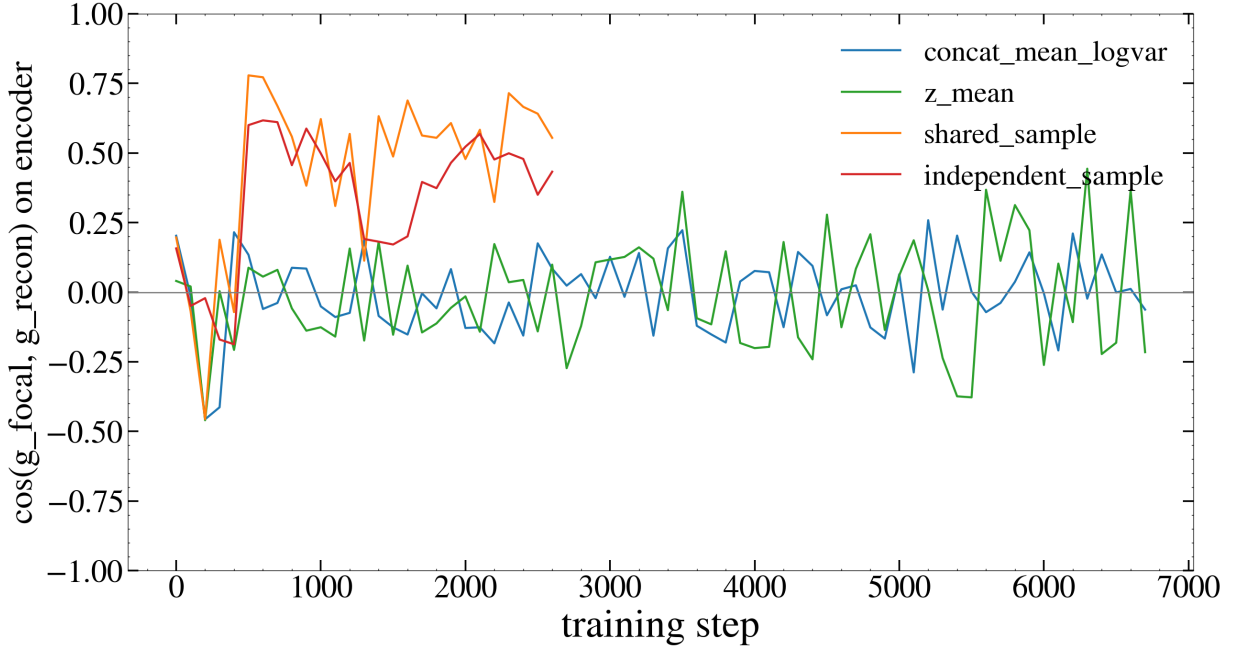


FIG. 2: Cosine similarity between the classifier (focal-loss) and the reconstruction gradient on the encoder weights, as a function of training step, for the four ablation modes. The deterministic modes hover around zero, while both sampling modes show a persistently positive alignment, indicating cooperative gradients once the reparameterisation noise enters the classifier path.

The comparison of `shared_sample` with `independent_sample` therefore answers whether the shared realisation or merely the presence of noise injection is responsible, while the deterministic modes (`concat_mean_logvar` and `z_mean`) serve as controls for whether the head requires the sampling operation at all.

To quantify the interaction, the cosine similarity between the classifier (binary-focal-loss) gradient and the reconstruction gradient, both taken with respect to the encoder weights, was recorded during training. The result is shown in Fig. 2.

We observe that the two deterministic modes (`concat_mean_logvar` and `z_mean`) fluctuate around zero throughout training, i.e. on average there is no consistent alignment between the classifier-induced and reconstruction-induced encoder gradients. In contrast, both sampling modes (`shared_sample` and `independent_sample`) exhibit a positive cosine similarity of order 0.5, indicating that the classifier and the decoder pull the shared encoder in a consistent direction. Since the effect is present for the independent sample as well, and is only marginally stronger for the shared sample, I conclude that the dominant ingredient is the injection of the reparameterisation noise into the classifier path rather than the sharing of the specific noise realisation itself. This provides direct evidence that, in the sampling regime, classifying on the latent and reconstruction of the signal are cooperative rather than antagonistic objectives on the encoder.

B. Loss design

The total VAE-classifier loss was constructed as a weighted sum

$$\mathcal{L}_{\text{total}} = \lambda_{\text{BFL}} \mathcal{L}_{\text{BFL}} + \lambda_{\text{tail}} \mathcal{L}_{\text{tail}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{corr}} \mathcal{L}_{\text{corr}}, \quad (2)$$

each term of which is explained below.

a. Classification and tail suppression. As in the preceding work [4], the classification term $\mathcal{L}_{\text{BFL}}$ is a binary focal cross-entropy [11] on the raw logit, which down-weights easy negatives and thereby focuses learning on the hard-to-classify windows that dominate the false-positive tail. To attack that tail more directly, an additional logsumexp tail loss is applied to the negative-class logits s_i^- ,

$$\mathcal{L}_{\text{tail}} = \frac{1}{\beta} \log \left[\frac{1}{N_-} \sum_{i=1}^{N_-} \exp(\beta \text{softplus}(s_i^- - m)) \right], \quad (3)$$

where the margin m sets where a negative logit begins to be penalised and β controls how sharply the loss concentrates on the largest outliers. The softplus function is defined as in [12]. This term targets precisely the failure mode in which a small number of noise windows force

the detection threshold upward, which, in fact, can aid in decreasing the needed SNR for "successful detection / classification"³.

b. Clean-signal reconstruction. The reconstruction target is set to be the stored clean signal rather than the raw input. The reconstruction loss is the mean-squared error between the decoded signal $\hat{s}$ and the clean target s , saved during preprocessing, averaged over the batch, the two quadratures and the window samples,

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \frac{1}{2T} \sum_{q \in \{I, Q\}} \sum_{t=0}^{T-1} \left(s_n^{(q)}(t) - \hat{s}_n^{(q)}(t) \right)^2, \quad (4)$$

where T is the window length, $\mathcal{B}$ is the batch size, and $q \in \{I, Q\}$ runs over the two quadratures.

The clean target is the trimmed injected waveform on its support and zero elsewhere, with the trimming condition $|s(t)| \geq 10^{-2} \max_t |s(t)|$, i.e. only samples down to $10^{-2} \max_t |s(t)|$ are kept⁴.

The zero baseline is valid because z-score normalisation makes the noise mean vanish. The clean-signal mode is used deliberately: it forces the decoder, and hence the latent, to encode the injected signal component rather than merely the mean noise statistics, which is what can make the latent a denoised summary rather than a faithful copy of the noisy input. However, I did not explicitly check whether the latent dimensions correspond to specific features of a signal itself. This is a possible direction that can be explored in the future.

c. Shape correlation. For clean-signal reconstruction, a correlation term is added on signal windows,⁵

$$\mathcal{L}_{\text{corr}} = 1 - \frac{\langle \hat{s} - \bar{\hat{s}}, s - \bar{s} \rangle}{\|\hat{s} - \bar{\hat{s}}\| \|s - \bar{s}\| + \epsilon}, \quad (5)$$

which constrains the reconstructed *shape* while the MSE term constrains the amplitude as well as the global phase. The correlation weight is kept small (~ 0.05), in particular early in training, because the correlation is invariant to a global amplitude scale and therefore cannot by itself fix the reconstructed amplitude (this is what the MSE term drives).

C. Train and Fine Tune

Training proceeds in two stages: a coarse pre-training stage followed by a fine-tuning stage. Both model variants, with and without the reconstruction branch, follow

the identical two-stage recipe, so that the reconstruction branch is the only *deliberate* difference between them. It should be noted, however, that this does not isolate the branch's effect completely: adding the branch alters the parameter count and the gradients on the shared components, so the two variants follow different optimisation trajectories, and part of any observed performance gap may stem from ordinary run-to-run stochasticity (initialisation, mini-batch sampling and the optimiser path) rather than from the branch itself. A fully rigorous attribution would require repeating each variant over several random seeds and comparing the mean performance gap against this run-to-run variance; here the comparison is therefore interpreted as indicative rather than conclusive.

a. Pre-training. The model is first trained from scratch on a broad, coarsely spaced range of injected signal-to-noise ratios (peak SNRs spanning 0.8–7.0), such that it is exposed to both "easy" high-SNR and moderately hard windows and learns a robust, well-organised latent. This stage uses a comparatively larger batch size (1024) and learning rate (5×10^{-4}) for fast, stable optimisation, a long linear KL warm-up, over 25 epochs, that gradually turns on the geometric latent regularisation, and only modest correlation- and tail-loss weights, whose role at this point is mainly to shape the reconstruction rather than to aggressively suppress the false-positive tail. The whole summary of used hyperparameters (HPs) can be found in Sec. A of the appendix.

b. Fine-tuning. The best pre-trained checkpoint (measured as the one with the absolute lowest total loss, shown in Eq. (2)) is then continued with the architecture held fixed while the training distribution and several loss weights are adjusted to concentrate on the hard, low-SNR regime that dominates the achievable false-positive rate.

The injected SNRs are shifted to lower values and sampled more finely and densely (0.2–3.5), and the number of injected signals per segment is doubled, giving denser coverage of the difficult examples. The batch size is halved, deliberately increasing the gradient noise to help the optimiser settle into a better optimum, and the learning rate is lowered for careful refinement. The correlation and tail-loss weights are increased by an order of magnitude to sharpen the reconstructed signal shape and to push harder on the false-positive tail, while the KL term is effectively switched off (its target weight reduced by two orders of magnitude, with no warm-up), as further regularization was not found to be beneficial. Early stopping is applied in both stages, with a slightly larger patience during fine-tuning. The exact numerical values for the fine tuning stage is collected in Appendix A.

³ "Successful detection" refers to the output logit of the latent classifier being above the set threshold for the desired false positives per year.

⁴ This was found to be sufficient, as the signal envelope decays exponentially and not all points will alter the noise sufficiently to aid in detection, just like a real signal is only detectable within a specific region around its maximum amplitude.

⁵ $\epsilon \ll \|\hat{s} - \bar{\hat{s}}\| \|s - \bar{s}\|$ is a "small" number that prevents division by zero.

V. DETECTION METRIC AND THRESHOLD CALIBRATION

The primary detection metric is retained from the preceding work [4] : the detection efficiency

$$\epsilon(\text{SNR}_{\text{peak}}) = \frac{N_{\text{detected}}}{N_{\text{injected}}}, \quad (6)$$

evaluated at a fixed window-based false-positive rate, where the injected SNR is the peak-amplitude SNR $\text{SNR}_{\text{peak}} = \max_t |s(t)|/\sigma_{\text{noise}}$. A signal is counted as detected if at least one of its associated windows exceeds a calibrated decision threshold (= event-based detection).

The decision threshold is calibrated to a target false-positive rate as follows: The model is run on noise-only windows and the false-positive rate is recorded as a function of the threshold; converting the window counts to a yearly rate through the number of windows processed per year yields the false positives per year at each threshold. Because the available data is far smaller than one year, the low-rate tail of this relation is fitted by a straight line in log-space, $\ln(\text{FPR}) = bt + c$, and inverted to obtain the threshold at a given target rate. An example of such a fit can be found in Appendix Sec. B of [4] .

Rather than fixing a single operating point, the threshold is calibrated across a range of targets, from 0.1 to 100 false positives per year, so that the required SNR can be studied as a function of the allowed false-alarm rate and study the scaling behaviour of each model.

It is important to note that although the VAE latent classifier model was trained using a sampling operation from the latent space, during inference the raw **z_mean** vector is fed into both the classifier and the decoder to make the model output deterministic rather than obtaining different distributions when performing one and the same forward-pass several times. The sampling operation is useful when the generative nature of a VAE is needed, i.e. obtaining new instances of data based on the learned latent. This is, however, not needed here.

The detection efficiency is then measured empirically as follows: For a given PBH mass and injected peak SNR, signals are injected into a separate evaluation time-series, distinct from the training, validation and noise-only calibration data, using the no-overlap injection scheme, so that no accidental signal overlap occurs. This is repeated over $N_{\text{runs}} = 1000$ independent injection realisations to obtain a statistically robust efficiency estimate at that operating point. Sweeping the injected peak SNR (from ≈ 0.1 to 9.6) and the PBH mass ($10^{-12} M_{\odot} \leq m_{\text{PBH}} \leq 10^{-6} M_{\odot}$, seven values) yields the characteristic S-shaped efficiency curves $\epsilon(\text{SNR}_{\text{peak}})$ (see Appendix C of the previous report [4] for examples); the error on each efficiency point is a Wilson-95 confidence interval.

Each efficiency curve is then fitted with a sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-k(x-x_0)}}, \quad (7)$$

and the SNR at which the efficiency is 0.95 is extracted from the fit; its uncertainty is propagated from the covariance matrix of the fitted parameters k and x_0 . Because the decision threshold itself depends on the target false-positive rate (see the calibration described above), this efficiency-curve fit and SNR_{95} extraction is carried out for *every* target rate in the scan, from 0.1 to 100 false positives per year. The final result is therefore the required SNR_{95} as a function of the allowed false-positive rate for each PBH mass, shown in Fig. 3, which shows how each model scales as more false alarms are tolerated.

It is important to mention that two classes of uncertainty enter the detection metric.

1. The **statistical uncertainty** on each efficiency point is estimated from Wilson-95 confidence intervals [13] . This is appropriate because each injection constitutes an independent Bernoulli trial (detected or not), so the number of detections follows a binomial distribution; the Wilson interval captures the asymmetry of the binomial error near efficiencies of 0 and 1 more faithfully than a Wald interval and remains well behaved in the low-count regime. These per-point errors are propagated into the extracted SNR_{95} through the covariance matrix of the fitted sigmoid parameters k and x_0 .
2. The second source of uncertainty is the **extrapolation uncertainty** of the false-positive tail used to set the threshold. Because the available dataset is orders of magnitude smaller than one year of data (roughly four to five orders of magnitude too small), the threshold corresponding to a low target rate cannot be measured directly: the log-space tail of the false-positive rate must be extrapolated over several orders of magnitude in false-positive fraction. Such a long extrapolation is intrinsically uncertain and the error inflates the further the tail is extended beyond the measured points. This uncertainty is not accounted for in the efficiency results, which I flag as one of the limitations of this work. The exact quantification of the size of this error is advisable for future works.

VI. DETECTION PERFORMANCE

I quantify the detection performance by the injected peak SNR required to reach a detection efficiency of 95%, denoted by SNR_{95} . As described in Sec. V , the decision threshold was recalibrated for every target false-positive (FP) rate between 0.1 and 100 FP/year.

A. Dependence on the false-positive operating point

For each PBH mass and target FP rate, I fitted the detection efficiency as a function of injected peak SNR

with a sigmoid function, shown in Eq. (7). Inverting that sigmoid at $\sigma = 0.95$ gives

$$\text{SNR}_{95} = x_0 + \frac{\ln 19}{k}, \quad (8)$$

where x_0 is the efficiency midpoint and k is the turn-on sharpness. From Eq. (8), we see that x_0 fixes the location of the transition, whereas k determines how rapidly the efficiency rises with SNR. To describe the dependence of SNR_{95} on the operating point, meaning the allowed FPs per year, I define

$$u = \log_{10} \left(\frac{R_{\text{FP}}}{1 \text{ yr}^{-1}} \right), \quad (9)$$

where R_{FP} is the target window-based false-positive rate (see Sec. 3 of [4] for further details on the calculation). Then I use a decreasing four-parameter logistic, shown in Eq. (10), to fit the dependence of SNR_{95} on the allowed FPs per year.

$$S_{95}(u) = S_{\text{lo}} + \frac{S_{\text{hi}} - S_{\text{lo}}}{1 + \exp[\kappa(u - u_0)]}. \quad (10)$$

Here, S_{hi} and S_{lo} are the high- and low-FP plateaus respectively, u_0 is the center of the transition, and κ sets its steepness in $\log_{10}(R_{\text{FP}})$. When the bend of Eq. (10) runs outside the measured range, I instead use

$$S_{95}(u) = a + bu, \quad b < 0, \quad (11)$$

which describes the observed interval without imposing unresolved plateaus. However, I note that this should be kept in mind when interpreting the scaling behaviour for the respective masses for higher and lower allowed FPs per year. Table I summarises these choices and the corresponding best-fit parameters are listed in Appendix B. The final result plot for both architectures is shown in Fig. 3.

In both panels of Fig. 3, we see that increasing the allowed FP rate lowers the required SNR. This is an expected operating-point trade-off: a more permissive threshold admits weaker signals along with more noise outliers. However, the effect is not uniform across architectures. The latent-only curves generally decline gently across the measured interval. For 10^{-11} , 10^{-10} , 10^{-9} , and $10^{-8} M_{\odot}$, their fitted logistic transition u_0 moves to an unphysical scale of order 10^{40} to 10^{43} FP/year and becomes indistinguishable from a straight line over the data. I therefore retain only the linear-in- u description for those masses. In contrast, the decoder-into-classifier curves exhibit a clearer high-SNR plateau, transition, and low-SNR plateau, and are statistically compatible with Eq. (10) for every mass⁶.

⁶ It should be noted that as the fitted logistic curve of Eq. (10) goes through most points with small residuals, the reduced χ^2 (denoted as χ^2_{ν}) decreases below one, with a minimal value of 1×10^{-3} at a mass of $1 \times 10^{-12} M_{\odot}$. In addition, small relative errors amplify non-zero residuals, which cause slightly elevated χ^2_{ν} of up to 2.9 for some fits.

The dashed curves make the model dependence of any long extrapolation explicit. All decoder-into-classifier curves saturate because Eq. (10) contains two asymptotes. The four latent-only curves listed above continue to scale linearly in $\log_{10}(R_{\text{FP}})$ because Eq. (11) contains no plateau. However, I do not interpret the linear scaling as evidence that one architecture physically saturates while the other does not. Measurements over a wider FP range would be required to distinguish persistent scaling from saturation. The curves can nevertheless provide a common operating-point parametrisation of the ML model for a future comparison with matched filtering, provided that the comparison uses the same window definition and the FP calibration and is not driven by the unvalidated extrapolated region.

The $10^{-10} M_{\odot}$ mass is the hardest to detect by a wide margin. At 1 FP/year the decoder-into-classifier model requires $\text{SNR}_{95} = 32.8(3)$ and the latent-only model $\text{SNR}_{95} = 18.3(5)$, roughly four to ten times the values needed at any other mass; relaxing the threshold to 100 FP/year lowers these to 30.2(4) and 17.0(3). Both architectures, therefore, do detect this mass, but only at large injected SNR relative to the needed one for other masses. The origin of this maximum is discussed quantitatively in Sec. VIB. I note, as the networks were trained solely on $10^{-8} M_{\odot}$ injections; adding $10^{-10} M_{\odot}$ signals to the training set is a natural way to test whether the required SNR at this mass can be reduced.

In summary, the latent-only architecture exhibits lower needed SNRs across the board as well as more favourable scaling beyond the measured range.

B. Dependence on PBH mass

The exact needed SNR values for 95% detection efficiency are listed in Appendix C.

Figure 4 reorganises the same measurements, discussed above, as a sensitivity profile over PBH mass at two representative operating points (1 FPs per year and 100 FPs per year). In addition to the factor-of-ten coarse scan across masses from $1 \times 10^{-12} M_{\odot}$ to $1 \times 10^{-6} M_{\odot}$, I sampled the region $10^{-11.5}$ – $10^{-9} M_{\odot}$ on a finer grid of 0.25 decades in order to resolve the shape around the peak. Because this finer scan was intended as an exploratory probe, each (mass, SNR) point was evaluated with 100 injection realisations rather than the 1000 used for the coarse scan, so its per-point statistical uncertainty is correspondingly larger.

To describe the pronounced maximum of Fig. 4, I use a single Lorentzian resonance profile on a constant heavy-mass floor,

$$S_{95}(x) = C + \frac{A}{1 + [(x - x_0)/\gamma]^2}, \quad x = \log_{10} \left(\frac{m}{M_{\odot}} \right), \quad (12)$$

where C is the heavy-mass floor, A the peak amplitude above the floor, x_0 the peak centre, and γ the

TABLE I: Empirical models used for the dependence of SNR_{95} on the target false-positive rate. The quoted reduced χ^2 (denoted as χ_ν^2) ranges refer to fits over the measured interval 0.1–100 FP/year.

Architecture	PBH masses [$M_\odot$]	Model in $u = \log_{10}(R_{\text{FP}}/\text{yr}^{-1})$	χ_ν^2
Decoder in classifier	all masses	decreasing logistic, Eq. (10)	0.002–2.9
Latent only	$10^{-12}, 10^{-7}, 10^{-6}$	decreasing logistic, Eq. (10)	0.01–0.30
Latent only	$10^{-11}, 10^{-10}, 10^{-9}, 10^{-8}$	linear in $\log_{10}(R_{\text{FP}})$, Eq. (11)	0.01–0.30

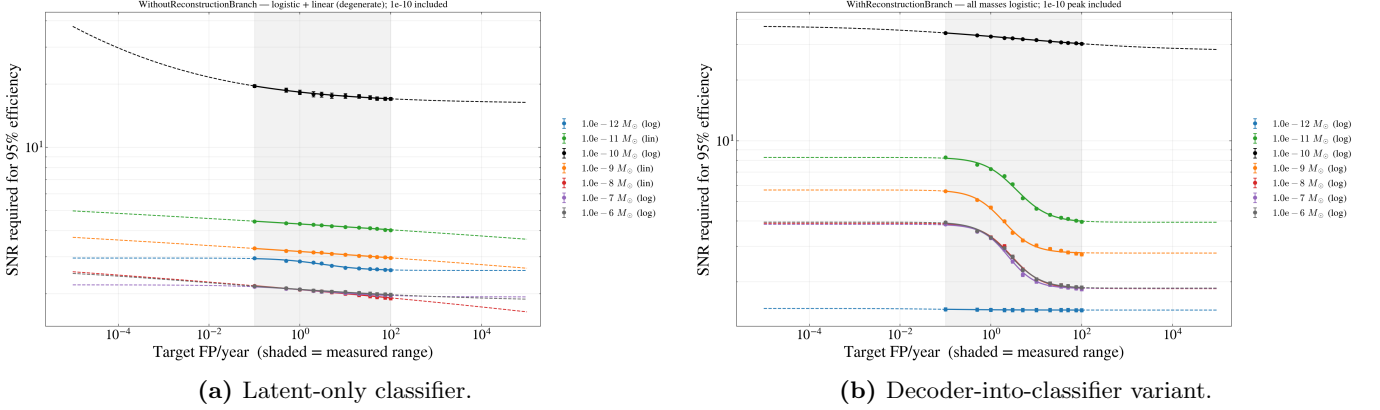


FIG. 3: Injected peak SNR required for 95% event-detection efficiency as a function of the target window-based false-positive rate. Markers and error bars show the measured SNR_{95} values and the uncertainties propagated from the efficiency-sigmoid covariance as described in Sec. V. The grey region marks the scanned interval, 0.1–100 FP/year. Solid curves show fits within that interval; dashed curves show their extrapolation to 10^{-5} – 10^5 FP/year. Equation (10) is used unless the logistic bend is unconstrained, in which case the linear form Eq. (11) is used. The $10^{-10} M_\odot$ curve (black) sits well above all other masses, reflecting the much higher SNR required at that mass.

half-width (in decades). Equation (12) is used purely as a visual shape descriptor and is not interpreted as a physical line profile. In addition, I also tested asymmetric (split) Lorentzians, gamma- and exponentially-modified-Gaussian profiles, and envelope-modulated oscillatory models; none of them reproduces the point-to-point behaviour. The best oscillatory description drives its period down to the sampling step itself, i.e. to the Nyquist limit of the 0.25-decade grid, which is the signature of an unresolved structure rather than a fitted feature. I therefore conclude that, at the present sampling, *no family of functions describes the mass profile statistically adequately*, and I show only the spike component to mark the one robust feature, the resonance at $10^{-10} M_\odot$. Establishing whether the surrounding oscillation is physical or an artefact of the mass sampling requires a finer scan (of order 0.05–0.1 decades) through 10^{-11} – $10^{-9} M_\odot$, ideally repeated with the full 1000 injection realisations per point so that the reduced statistics of the present exploratory grid can be excluded as a contributing factor.

Across the four series in Fig. 4, we see the same broad structure: the required SNR is low ($\text{SNR}_{95} \approx 2$ –4) at the light and heavy ends and rises to a narrow, pronounced maximum at $10^{-10} M_\odot$. At 1 FP/year the decoder-into-classifier model peaks at $\text{SNR}_{95} = 32.83(28)$ ($10^{-10} M_\odot$)

and falls to approximately 3.30 at 10^{-8} – $10^{-6} M_\odot$; the latent-only model peaks at 18.26(45) and falls to approximately 2.10.

The finer scan, however, reveals that the transition between the floor and the peak is not a smooth rise. The measured SNR_{95} values jump from point to point across $10^{-11.5}$ – $10^{-9} M_\odot$ by amounts much larger ($\gg 3\sigma$) than their statistical uncertainties, and the same jagged pattern appears at every FP operating point. As a consequence, the single-Lorentzian spike of Eq. (12) fits the data statistically poorly: even though it pins the resonance centre at $x_0 \simeq -10.05$ (see Tab. IX for errors) to within a few thousandths of a decade, the reduced chi-squares lie between 77 and 315, with individual residuals reaching several tens of sigma.

Interestingly, the location of the maximum coincides with a simple time-scale matching in the signal simulation. The injected chirp sweeps across the waveform frequency span $\Delta f = f_s = 14 \text{ MHz}$ (f_s is the sampling frequency) at the constant chirp rate $k(m_{\text{PBH}})$ [6], depicted in Eq. (13).

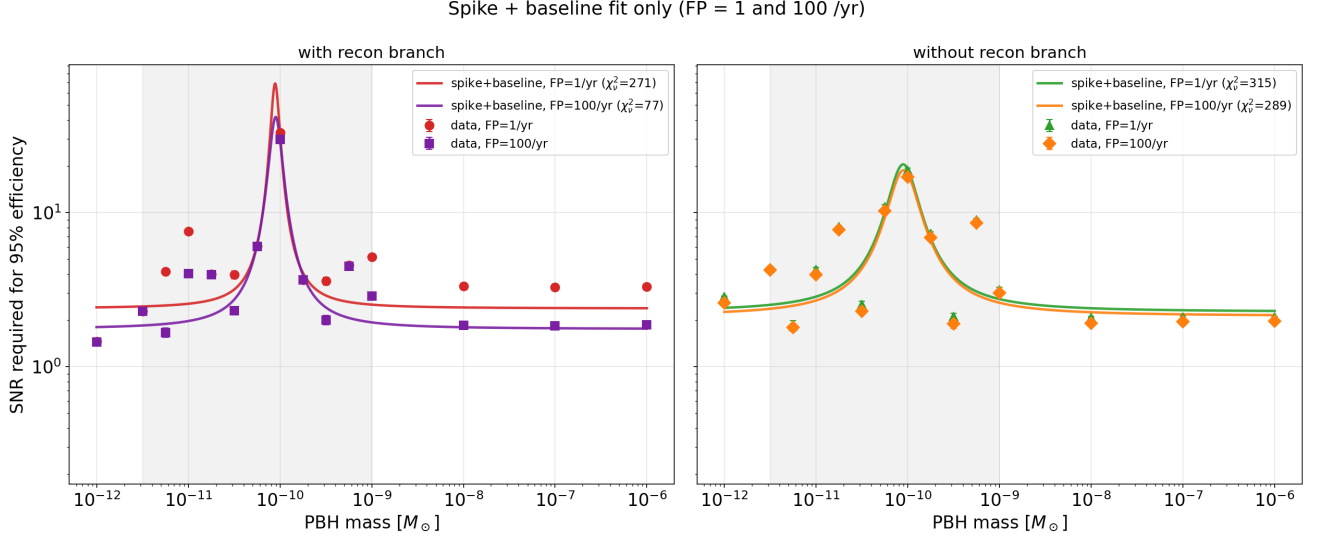


FIG. 4: Mass dependence of the injected peak SNR required for 95% detection efficiency. Markers show results for the latent-only (left) and decoder-into-classifier (right) variants at target rates of 1 and 100 FP/year, including the finer 0.25-decade scan across $10^{-11.5}$ – $10^{-9} M_{\odot}$; error bars contain only the uncertainty propagated from the corresponding efficiency-sigmoid fit. The solid curves show the single-Lorentzian spike of Eq. (12), fitted independently at each operating point. The fit does *not* describe the data statistically well (χ^2_{ν} between 77 and 315): it correctly locates the narrow resonance at $x_0 \simeq -10.05$ ($m \simeq 10^{-10} M_{\odot}$, half-width $\gamma \lesssim 0.18$ dec), but the measured SNR_{95} values around that resonance oscillate strongly from one mass point to the next by amounts far exceeding their error bars. Only the spike component is drawn. The error bars of most data points are barely visible using the logarithmic y -axis scaling. This scaling, however, better shows the difference between the operating points curves.

$$k(m_{\text{PBH}}) = k_0 \left(\frac{m_{\text{PBH}}}{10^{-9} M_{\odot}} \right)^{5/3} \left(\frac{f_{\text{start}}}{1 \text{ GHz}} \right)^{11/3},$$

$$k_0 = 4.62 \times 10^{11} \text{ Hz s}^{-1}, \quad (13)$$

where m_{PBH} is the PBH mass, and $f_{\text{start}} = f_0 - f_{\text{span}}/2$ is the lower-edge (start) frequency of the waveform grid. Here $f_0 = 5 \text{ GHz}$ (so $f_{\text{start}} = f_0 - f_s/2 = 4.993 \text{ GHz}$).

The duration a signal is inside the resonance band of the cavity is (see Sec. 4.3 of [7] for further details):

$$\Delta t(m_{\text{PBH}}) = \frac{\Delta f}{k(m_{\text{PBH}})}. \quad (14)$$

The cavity responds on its ring-down time scale ⁷

⁷ Here $\tau = 1/(\pi\Gamma)$ is the cavity's *field* (amplitude) ring-down time, i.e. the $1/e$ decay time of the voltage transient. It is twice the photon lifetime (energy decay time) $\tau_{\text{ph}} = 1/(2\pi\Gamma)$, which satisfies the Fourier relation $\tau_{\text{ph}}\Gamma = 1/(2\pi)$: since the stored energy scales as the amplitude squared, the field decays half as fast as the energy. The field time is the self-consistent choice here because the single-pole cavity response $H(f) = [(f - f_0) + i\Gamma/2]^{-1}$ used in the simulation has the impulse response $h(t) \propto e^{-\pi\Gamma t}$, so the readout voltage rings down with $\tau = 1/(\pi\Gamma)$.

$$\tau = \frac{1}{\pi\Gamma_0} \approx 3.18 \mu\text{s}, \quad (15)$$

with $\Gamma_0 = 100 \text{ kHz}$ the used cavity's FWHM in the signal simulation.

Defining a "cross-over mass" $m_{\star}$, defined as the PBH mass needed to create a chirp $k(m_{\star})$ that results in $\Delta t(m_{\star}) = \tau$, for $m_{\text{PBH}} \ll m_{\star}$ the signal is long compared with the cavity response ($\Delta t \gg \tau$); for $m_{\text{PBH}} \gg m_{\star}$ it is short ($\Delta t \ll \tau$). Setting $\Delta t = \tau$ selects the crossover mass $m_{\star}$:

$$m_{\star} \simeq 1.124 \times 10^{-10} M_{\odot}, \quad (16)$$

where the injected transient is least distinguishable from the cavity's own response. The finer mass scan now samples this region directly, and the measured SNR_{95} maximum sits at $m \simeq 10^{-10} M_{\odot}$, in agreement with $m_{\star}$: there Eqs. (14)–(15) give $\Delta t \approx 3.9 \mu\text{s} \approx \tau$. The fitted spike centre x_0 of Eq. (12) likewise falls at $x_0 \simeq -10.05$ for all four series, consistent with $m_{\star}$ to within the 0.25-decade sampling. Resolving the exact peak position and the coincidence with $m_{\star}$ more sharply would require the finer scan mentioned above.

C. Limits on a mechanistic interpretation

The performed measurements establish where the classifier detects the simulated signals, but they do not identify which waveform feature drives the decision. Because the classifier acts only on the latent code, the discrimination that matters is the one realised there: how far the signal and noise windows separate in latent space is, by construction, what the classifier has to work with. A principal-component view of the latent along its highest-variance directions is a useful visual probe of that separation, although here it is limited by the absence of a dedicated false-positive (noise-only) set in that view.

A plausible, though unverified, detection mechanism here is the progressive growth of the effective receptive field through the convolutional stack, followed by the storage of the integrated and compressed information in the latent space. The present analysis does not demonstrate that mechanism explicitly, as this was not studied in detail. The network could instead rely mainly on a shorter, high-amplitude part of the waveform, in which case a receptive field covering the full signal would not be necessary. I therefore do not identify a specific detection mechanism in this report. To assess whether the receptive field is the key to detection, a controlled receptive-field ablation, together with input signal parameter perturbations and latent-feature attribution, is a possible way to crystallise the underlying detection mechanism.

Beyond these features, and based on the results of Sec. VIB, a possible working hypothesis is that the network encodes timescale information tied to the cavity response. Such a timescale would directly control the overlap of the signal and noise distributions in the latent space and would thereby play a key role in detection from the latent code. As a brief, exploratory check of exactly this overlap, I asked whether the detection difficulty is already visible in the latent representation that the classifier actually reads, rather than only in the final decision. For each PBH mass I encoded a set of injected-signal windows and a set of noise windows at the physical cavity FWHM $\Gamma_0 = 100$ kHz, and measured how far the two clouds separate along the most discriminative direction of the 32-dimensional latent space. Projecting the latent vectors onto that direction and denoting the resulting signal and noise means and variances by (μ_s, σ_s^2) and (μ_n, σ_n^2) , the separation is quantified by the standardised distance

$$d' = \frac{|\mu_s - \mu_n|}{\sqrt{\frac{1}{2}(\sigma_s^2 + \sigma_n^2)}}. \quad (17)$$

Equation (17) measures the gap between the signal and noise latent clouds in units of their average within-class spread: $d' \rightarrow 0$ means the encoder maps signal and noise on top of one another (they are indistinguishable in the latent code), whereas a large d' means the two are cleanly separated. Figure 5 shows the result: for both architectures the latent separability collapses to a sharp min-

imum at $10^{-10} M_\odot$ - from $d' \simeq 3$ –3.5 away from the peak down to $d' \lesssim 1$ at the resonance - coinciding with the SNR_{95} maximum of Fig. 4 and with the $\Delta t = \tau$ crossover. In other words, the hardest-to-detect mass is also the mass at which the encoder can barely tell signal from noise in its latent code, giving direct, if preliminary, evidence that the signal/noise overlap in the latent space is governed by the same $\Delta t = \tau$ condition. I stress that this is a quick, exploratory study only: the analysis script was fully written with Claude Opus 4.8 [14], the separability is estimated with a simple per-mass linear discriminant, and no attempt has been made to control for the many confounds (window statistics, latent dimensionality, the absence of a dedicated false-positive set in this view). The result is therefore indicative rather than conclusive, but the clean coincidence of the latent-space minimum with the detection-regime peak is striking and certainly worth investigating further. Additionally, also during model testing using principal component analysis consistently yielded visually highly overlapping noise and signal-plus-noise distributions in the latent space.

This still does not explain which features are extracted from the raw input, nor how the model tells noise-only from signal-plus-noise windows, which is especially interesting at low SNR where the presence of a signal is not apparent to the bare eye. The above mentioned learned time scale hypothesis is, however, directly testable: one repeats the full training, fine-tuning and analysis of this report while choosing a different cavity FWHM Γ for the bandwidth response. If the peak of Fig. 4 shifts to the new crossover mass $m_\star$ predicted by the altered $\tau = 1/(\pi\Gamma)$, one can conclude that the model has learned a timescale specific to the cavity.

This matters before deploying such a model across several cavities of differing Γ , since the model could then behave differently at the same masses, that is, the detection efficiency would change from cavity to cavity.

VII. COMPARISON OF THE LATENT-ONLY AND DECODER-INTO-CLASSIFIER ARCHITECTURES

The two networks use the same data pipeline and two-stage training recipe, discussed in Sec. IV C, and differ deliberately only in whether detached decoder features are concatenated with the latent features at the classifier input. I compare them through a taken ratio

$$\mathcal{R}_{95} = \frac{\text{SNR}_{95}^{\text{decoder branch}}}{\text{SNR}_{95}^{\text{latent only}}}. \quad (18)$$

Thus, $\mathcal{R}_{95} < 1$ means the model with the decoder branch is more sensitive, whereas $\mathcal{R}_{95} > 1$ means the latent-only classifier performs better, and $\mathcal{R}_{95} = 1$ denotes equal sensitivity.

Figure 6 shows that the decoder branch does not provide a uniform advantage. For 10^{-11} , 10^{-10} , and 10^{-9} – $10^{-6} M_\odot$, we see $\mathcal{R}_{95} \simeq 1.7$ –1.9 at the strictest operating

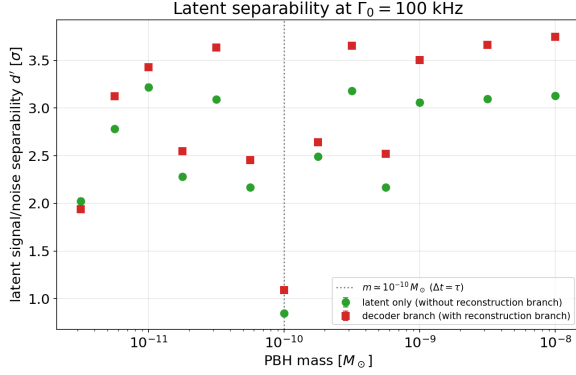


FIG. 5: Latent signal/noise separability versus PBH mass at a FWHM of $\Gamma_0 = 100$ kHz. Markers show d' (Eq. (17)), the standardised separation between the signal and noise latent clouds along the Fisher direction, with 1σ error bars; the dotted line marks $10^{-10} M_\odot$ ($\Delta t = \tau$). Both architectures show a pronounced minimum there, where the encoder can least distinguish signal from noise. These are exploratory result as discussed in Sec. VIC.

TABLE II: Detection comparison at a target rate of one false positive per year. The quoted SNR_{95} errors come from the efficiency-fit covariance, and the ratio uncertainty is propagated in quadrature.

$m [M_\odot]$	$\text{SNR}_{95}^{\text{decoder branch}}$	$\text{SNR}_{95}^{\text{latent only}}$	$\mathcal{R}_{95}$
10^{-12}	1.45(3)	2.85(3)	0.51(1)
10^{-11}	7.24(7)	4.31(4)	1.68(2)
10^{-10}	32.8(3)	18.3(4)	1.80(5)
10^{-9}	4.67(6)	3.17(4)	1.47(3)
10^{-8}	3.32(5)	2.09(3)	1.59(4)
10^{-7}	3.29(5)	2.10(3)	1.57(3)
10^{-6}	3.32(5)	2.10(3)	1.58(3)

point of 0.1 FP/year. The branch therefore requires 70–90% more injected peak SNR than the latent-only model in this regime. For the hardest resolved mass $10^{-10} M_\odot$ the disadvantage is essentially flat across the whole FP scan, $\mathcal{R}_{95} \simeq 1.78$ –1.80: the branch consistently needs about 1.8 times the SNR there. For the lighter and heavier masses the ratio decreases as the threshold is relaxed and reaches approximately 0.92–0.99 by 100 FP/year, where those architectures are close to break-even. The $10^{-12} M_\odot$ case is the clear exception. Across the entire FP scan we see $\mathcal{R}_{95} \simeq 0.49$ –0.56: the decoder-into-classifier model needs only about half the SNR of the latent-only model. Table II gives the comparison at the reference operating point of 1 FP/year and makes the sign change explicit: the branch is strongly favoured at $10^{-12} M_\odot$ ($\mathcal{R}_{95} = 0.51$), and is disfavoured at most operation points for every other mass, most strongly at a mass of $10^{-10} M_\odot$ where $\mathcal{R}_{95} = 1.80$.

Taken together, the ratio in Fig. 6 and Table II do not

support the claim that the reconstruction branch generally improves detection by providing an additional “view-point” of the same underlying latent code, which was the primary motivation of adding the reconstruction branch (see Sec. II).

Because each architecture was trained only once, I treat this attribution as indicative rather than conclusive. The branch changes the parameter count and the optimisation trajectory even though the training recipe is otherwise identical. Multiple independently seeded training runs could further be used to separate a reproducible architectural effect from run-to-run stochasticity. However, I would attribute this effort more towards one of academic interest rather than it being useful to improve detection.

Finally, I highlight that the used architectures, in this report, must not be inherently better-performant than the earlier CNN-LSTM classifier [4]. That model was evaluated with the previous 2 MHz frequency span, whereas the present networks use the 14 MHz detector-matched simulation. The signal duration and likely detection mechanism therefore differ, so a direct SNR comparison would confound the architecture with the simulation change. A fair baseline requires retraining the CNN-LSTM, on exactly the same injected waveforms, windowing, noise data, event definition, and FP-tail calibration used here.

VIII. A PROBABILISTIC PARAMETER-ESTIMATION PROTOTYPE

Lastly, I want to give a brief introduction into a possible prototype of signal parameter estimation on the latent space.

Motivated by the physics-shapeable nature of the latent space (Sec. II), the same latent representation that drives detection can, in principle, be reused to estimate physical signal parameters. These in turn can again shape the latent space, potentially increasing separation of signal and noise mapping and therefore aid the latent classifier in detection.

This section sketches how such an extension can be built and which design choices matter, based on a first exploratory prototype. It is intended as a starting point rather than a finished pipeline. The natural targets include the physical parameters that shape the injected waveform: the PBH mass m_{PBH} , which sets the chirp rate and hence the signal duration through Eq. (13), together with quantities such as the injected peak SNR (or the source distance-detector distance if an analytical formula is known) and the signal start time or phase within the analysis window.

Each such target can be attached as a small regression head on top of the shared latent features. Rather than a single point estimate, each head can predict a Gaussian predictive distribution $\mathcal{N}(\mu, \sigma^2)$ and be trained by the (β -weighted) negative log-likelihood [15], so that a

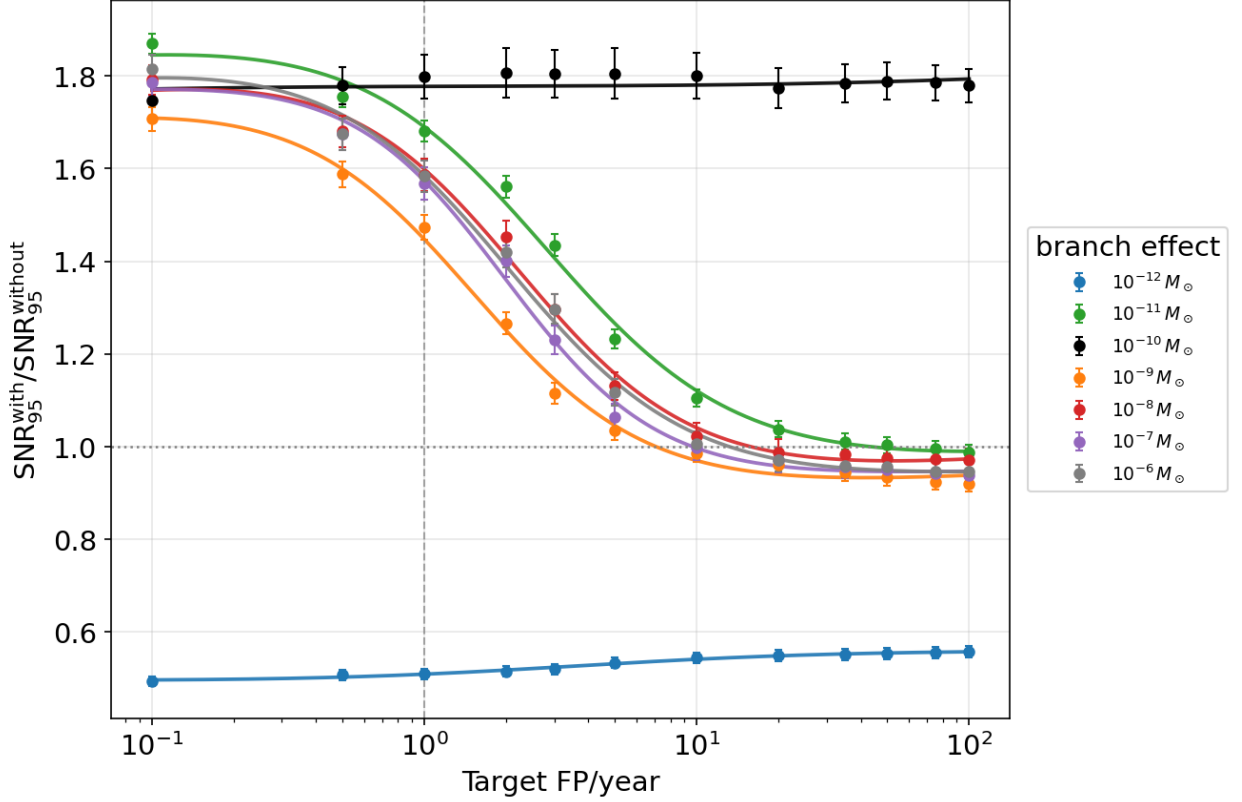


FIG. 6: Effect of exposing the classifier to the detached decoder reconstruction. The vertical axis shows the ratio $\mathcal{R}_{95} = \text{SNR}_{95}^{\text{decoder branch}} / \text{SNR}_{95}^{\text{latent only}}$ as a function of the target window-based FP rate. Markers are the measured ratios; solid curves are the ratio of the fitted $\text{SNR}_{95}(R_{\text{FP}})$ model curves, as done and explained in Sec. VIA, of each architecture. Error bars are obtained by propagating the two efficiency-fit uncertainties in quadrature. The dotted gray horizontal line marks equal performance.

calibrated uncertainty is reported alongside every prediction. The quality of that uncertainty can be checked with the probability-integral-transform (PIT) [16], whose histogram is flat for a correctly calibrated predictive distribution and therefore provides a direct diagnostic of over- or under-confidence.

To briefly test the above, I used Claude Opus 4.8 [14] to write a script that performs the aforementioned approach. When analysing the results, two things quickly emerged from the prototype that are worth mentioning:

1. Generalisation across PBH mass is poor unless a sufficiently dense set of intermediate masses is present in the training set: with too few distinct masses the model interpolates unreliably and the mass calibration degrades. The density of the training grid in parameter space is thus a primary lever for improving the estimator, and it should be chosen finer than the factor-of-ten grid (this grid was used in the brief testing of such a prototype).
2. The appropriateness of the Gaussian predictive is not uniform across the mass range, but is governed by the ratio of the signal duration to the cavity

ring-down time τ . For heavy masses ($\Delta t \ll \tau$), the chirp is far shorter than the cavity response, so the recorded waveform saturates to the cavity's own impulse response and becomes nearly mass-independent. There the true predictive is not a localised mass estimation but a broad, one-sided object (a bound rather than a value): a single Gaussian misrepresents its shape, and while the learned σ should inflate to flag the loss of information and increase uncertainty, the central estimate μ is untrustworthy. A natural remedy is to merge the mutually indistinguishable heavy masses into a single group and predict it as one class, replacing the ill-posed continuous regression above the identifiability threshold by a well-posed prediction of group membership (equivalently, a lower bound $m_{\text{PBH}} \gtrsim m_{\text{group}}$), while retaining the continuous Gaussian head only where the mass is identifiable ($\Delta t \gtrsim \tau$).

The so-found preliminary results are listed in Appendix D.

IX. SUMMARY

In this work I developed and studied a VAE-classifier that performs HFGW signal detection on a single, compressed latent representation, simultaneously shaped by a generative decoder and by physics-motivated loss terms (Sec. II, Fig. 1). The central design idea is architectural *locality*: unlike a general deep network, the VAE provides one well-defined place where all retained, compressed information about an input resides, which makes the representation inspectable and reusable. Two variants were compared throughout, a *latent-only* classifier and a *decoder-into-classifier* variant that additionally receives a stop-gradient reconstruction of the signal. Several results follow from this design:

A gradient-based ablation (Sec. IV A, Fig. 2) showed that the classifier acting on the latent and the decoder *cooperate* on the shared encoder once the reparameterisation noise is injected into the classifier path, and that this cooperation is driven by the noise injection itself rather than by sharing a specific noise realisation. Matching the waveform frequency span to the 14 MHz detector sampling rate (Sec. III) lengthened the signals by a factor of seven and moved detection out of the single-kernel template-matching regime; empirically this longer, detector-matched waveform *lowered* the SNR required for detection, which I attribute to the removal of a simulation–detector grid mismatch rather than to window occupancy alone.

Detection was quantified by SNR_{95} , the injected peak SNR needed for 95% event-detection efficiency (Sec. V), evaluated across FP rates from 0.1 to 100 per year and PBH masses from 10^{-12} to $10^{-6} M_{\odot}$ (Secs. VI–VII; the full SNR_{95} values are tabulated in Appendix C and the fit parameters in Appendix B).

The latent-only architecture needs lower SNR across the board and scales more gently as the tolerated FP rate is relaxed (Fig. 3, Table I). The mass profile is strongly non-monotonic, with a pronounced maximum at $10^{-10} M_{\odot}$ (Fig. 4); this mass coincides with the cavity time-scale matching $\Delta t = \tau$ (crossover mass $m_{\star} \simeq 1.12 \times 10^{-10} M_{\odot}$, Eq. (16)), where the injected transient is least distinguishable from the cavity’s own response. A finer scan revealed that the transition into this peak is jagged rather than smooth, and no simple family of functions describes it at the present sampling (Table IX); only the narrow resonance at $10^{-10} M_{\odot}$ is a robust feature.

Comparing the two variants (Sec. VII, Fig. 6 and Table II), the reconstruction branch does *not* provide a uniform advantage: it is strongly favoured only at $10^{-12} M_{\odot}$ and disfavoured at most other masses and operating points, so I do not find support for the original motivation that a second, signal-domain “view” of the same latent code generally improves detection. Because each variant was trained only once, this comparison is indicative rather than conclusive.

An exploratory look inside the latent space (Sec. VIC, Fig. 5) tied these observations together: the encoder’s

signal/noise separability d' collapses to a sharp minimum at exactly $10^{-10} M_{\odot}$, coinciding with the SNR_{95} maximum and the $\Delta t = \tau$ crossover. This gives preliminary evidence that the difficulty is already imprinted in the latent code and is consistent with the working hypothesis that the network encodes a cavity-specific time scale.

Finally, a probabilistic parameter-estimation prototype (Sec. VIII, with diagnostics in Appendix D, Figs. 7 and 8) demonstrated that the same latent can be turned towards physical parameter inference with calibrated Gaussian uncertainties, while also exposing where such an approach must be adapted: for heavy masses ($\Delta t \ll \tau$) the waveform saturates to the mass-independent cavity response, the mass becomes unidentifiable, and a single Gaussian should give way to a bound or a grouped-class prediction.

X. CONCLUSION

Overall, of the two variants the latent-only VAE-classifier is the stronger performer, requiring lower SNR for detection across the mass range than the decoder-into-classifier variant, which only overtakes it at the lightest mass $10^{-12} M_{\odot}$. Both, however, add what the previous CNN–LSTM classifier [4] did not have: a single, explicit, physically shapeable representation that is open to inspection and reuse.

I emphasise that no claim of superiority over the earlier CNN–LSTM classifier is made, since that model was evaluated on a physically different (2 MHz) waveform; a fair comparison would require retraining it on the identical simulation, windowing, noise data and FP calibration used here. I regard two issues as the dominant remaining bottlenecks: the false-positive tail, which the tail-suppression loss (Sec. IV B) mitigates but does not eliminate, and the extrapolation uncertainty in the threshold calibration (Sec. V), which cannot be removed without processing on the order of a year of data, but might be bettered when explicitly analysed.

XI. OUTLOOK

Several concrete directions follow, many of which were flagged throughout the report.

The single most valuable improvement is a further reduction of the false-positive tail, together with investigating the extrapolation uncertainty. Because the decision threshold is fixed entirely by the tolerated FP rate (Sec. V), suppressing the small number of noise outliers that push the threshold upward would allow the many signals that already produce intermediate logit values to cross it, directly raising the detection efficiency at fixed FP rate. Improving the tail is therefore, to first order, equivalent to improving detection; the tail-loss of Sec. IV B is one route, but other, here-unspecified outlier-rejection methods could be explored.

More broadly, the loss and training setup developed here (focal + tail + clean-signal reconstruction + correlation, Sec. IV B, Eq. (2)) together with the concept of architectural locality (see Sec. II) is a reusable recipe that need not be confined to a VAE.

A decoder head can be attached to essentially any classifier, for example the earlier CNN–LSTM model, so that the network drives signal reconstruction alongside classification, with the correlation and clean-MSE terms shaping that reconstruction. Such a head restores the architectural locality described in Sec. II and thereby also makes it possible to attach the parameter-estimation heads of Sec. VIII while trying a different architecture. It should be stressed that adding a decoder head does not by itself yield a VAE: the KL term, together with the sampling operation that lets the decoder generate previously unseen data, is the crucial ingredient that turns a plain autoencoder into a VAE and geometrically shapes the latent. Therefore, whether a architecture as described above is just a permutation of a VAE really depends on whether a KL loss is used or not.

Furthermore, whether the classifier and the attached

decoder gradients still align in such a hybrid remains an open question that can be answered with the same gradient-based ablation used here (Sec. IV A, Fig. 2).

The cavity-time-scale hypothesis is directly testable by repeating the full training, fine-tuning and analysis with a different cavity linewidth Γ : if the SNR_{95} peak of Fig. 4 shifts to the crossover mass predicted by the altered $\tau = 1/(\pi\Gamma)$, the network has learned a cavity-specific time scale, which matters before deploying one model across cavities of differing Γ .

Finally, the parameter-estimation prototype is a natural next step towards physics-informed inference on the latent.

XII. ACKNOWLEDGEMENTS

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Appendix A: Hyperparameters used during Pre-Training and Fine-Tuning

Tables III–VII list the hyperparameters for both training stages. The fine-tuning stage continues from the best pre-trained checkpoint with the network architecture and signal-simulation settings held fixed; only the data/injection distribution, several loss weights and the optimisation schedule differ. A dash (=) denotes a value unchanged from pre-training. Both model variants (with and without the reconstruction branch) use these settings.

TABLE III: Parameters used for the signal simulation and cavity response.

Parameter	Pre-training	Fine-tuning
Cavity resonance frequency f_0	5 GHz	=
Cavity linewidth Γ	100 kHz	=
Waveform length N_{gw}	32768	=
Cavity response model	real Lorentzian	=
Waveform grid mode	detector rate	=
Detector sampling rate f_s	14 MHz	=
Reject time-aliased waveforms	yes	=
Reject unrepresentable injections	yes	=

TABLE IV: Data preparation and injection settings.

Parameter	Pre-training	Fine-tuning
Input representation	I/Q (2 ch.)	=
Window size T	1024 samples	=
Step size	51 ($\approx T/20$)	=
Samples read per file	1.12×10^8	=
Normalisation	z -score	=
Injection mode	SNR-based, no-overlap	=
Injected peak-SNR grid	0.8–7.0	0.2–3.5
Training PBH mass	$10^{-8} M_\odot$	=
Signals/segment (train/val)	14000/2800	28000/5600
Train/val/test split	0.84/0.15/0.01	=
Random seed	42	=

TABLE V: Network architecture parameters. Convolutional parameters are per layer.

Component	Pre-training	Fine-tuning
Encoder filters	[16, 32, 64, 96, 96]	=
Encoder kernel sizes	[16, 7, 7, 7, 15]	=
Encoder strides	[2, 2, 4, 4, 1]	=
Decoder kernel sizes	[16, 7, 7, 7, 15]	=
Decoder output	real/imag (I/Q)	=
Latent dimension d	32	=
Activation (enc./dec.)	SiLU	=
Classifier hidden units	[64, 32]	=
Classifier dropout	0.1	=
Classifier latent input	sampled z	=
Recon. branch filters	[16, 32, 64, 96]	=
Recon. branch kernel sizes	[16, 8, 8, 8]	=
Recon. branch strides	[4, 4, 2, 1]	=
Recon. branch dense units	32	=
Recon. branch magnitude ch. yes		=

TABLE VI: Loss terms and weights (cf. Sec. IV B).

Term	Pre-training	Fine-tuning
Focal weight λ_{BFL}	10.0	=
Focal γ	4.0	=
Focal α	0.25	=
Tail weight λ_{tail}	0.05	0.5
Tail β	10.0	=
Tail margin m	−2.0	=
KL β_{KL} (start/end)	$10^{-5}/10^{-3}$	$10^{-5}/10^{-5}$
KL warm-up	25 epochs	0 epochs
Reconstruction weight λ_{rec}	5×10^{-3}	=
Reconstruction target	clean signal	=
Correlation weight λ_{corr}	0.05	0.5
Correlation type	phase-inv. (I/Q)	=
Correlation ϵ	10^{-8}	=

TABLE VII: Optimisation and in-training detection monitor values.

Parameter	Pre-training	Fine-tuning
Optimiser	AdamW	=
Learning rate	5×10^{-4}	1×10^{-4}
Weight decay	10^{-4}	=
Adam ($\beta_1, \beta_2, \epsilon$)	(0.9, 0.999, 10^{-7})	=
Gradient clip norm	1.0	=
Batch size	1024	512
Max epochs	100	=
Early stopping epochs	15	20
Frozen components	—	none
Detection target FP/year	1.0	=
Threshold sweep points	200	=
Log-fit tail fraction	10^{-4}	=
Min. tail points	4	=

Appendix B: Fit parameters

TABLE VIII: False-positive dependence of SNR_{95} , per mass. Parameters of the decreasing logistic Eq. (10), or of the linear fallback Eq. (11) where the logistic bend is unconstrained (rows marked †; there the first column gives the intercept a and the third the slope b). Uncertainties are 1σ from the fit covariance. Where the logistic is degenerate the corresponding parameters are only weakly constrained, and unconstrained entries are marked “–”.

Architecture	$m [M_\odot]$	S_{lo}	or a	S_{hi}	κ	or b	u_0
decoder in classifier	10^{-12}	1.447(35)		1.5(10)	–		–
	10^{-11}	3.941(40)		8.256(67)	2.81(15)		0.432(20)
	10^{-10}	27.8(76)		37(10)	0.6(12)		0.2(14)
	10^{-9}	2.772(23)		5.693(70)	3.09(20)		0.167(26)
	10^{-8}	1.847(22)		3.895(54)	3.04(24)		0.328(29)
	10^{-7}	1.852(19)		3.846(50)	3.43(26)		0.278(27)
	10^{-6}	1.859(23)		3.946(57)	2.95(23)		0.289(30)
latent only	10^{-12}	2.580(36)		2.958(52)	2.03(77)		0.47(16)
	10^{-11} †	4.300(16)		–	–0.133(14)		–
	10^{-10} †	18.47(15)		–	–0.81(11)		–
	10^{-9} †	3.179(15)		–	–0.107(12)		–
	10^{-8} †	2.091(12)		–	–0.091(10)		–
	10^{-7}	1.93(11)		2.20(15)	1.2(15)		0.32(62)
	10^{-6}	1.83(17)		–	–		–

TABLE IX: Mass-profile spike fit. Parameters of the single-Lorentzian spike on a constant floor, Eq. (12), fitted independently at each operating point to the merged coarse and fine (0.25-decade) mass scan. The resonance sits at $x_0 \simeq -10.05$ ($m \simeq 10^{-10} M_\odot$) with half-width $\gamma \lesssim 0.18$ dec in every case. The large χ_ν^2 reflects the strong, reproducible point-to-point oscillation of the measured SNR_{95} around the resonance, which no smooth family describes (see Sec. VIB); only the spike component is retained. Uncertainties are 1σ .

Architecture	FP/yr	C	A	x_0	γ	χ_ν^2
decoder in classifier	1	2.38(2)	66(18)	–10.053(5)	0.049(8)	271
	100	1.75(2)	40(4)	–10.047(4)	0.071(5)	77
latent only	1	2.27(2)	18.3(7)	–10.051(4)	0.172(5)	315
	100	2.13(2)	16.7(6)	–10.046(4)	0.178(5)	289

Appendix C: Tabulated SNR_{95} values

Tables X and XI list the injected peak SNR required for 95% event-detection efficiency, SNR_{95} , for every PBH mass and every scanned target false-positive rate, for the decoder-into-classifier and the latent-only architecture respectively. The quoted uncertainties (concise notation, value(unc)) are the 1σ errors propagated from the efficiency-sigmoid covariance; the FP-tail calibration systematic is not included.

TABLE X: SNR_{95} for the decoder-into-classifier architecture (with reconstruction branch).

FP/yr	10^{-12}	10^{-11}	10^{-10}	10^{-9}	10^{-8}	10^{-7}	10^{-6}
0.1	1.460(27)	8.267(59)	34.16(32)	5.624(56)	3.900(49)	3.853(47)	3.938(49)
0.5	1.454(27)	7.602(63)	33.30(28)	5.083(62)	3.557(53)	3.542(52)	3.560(52)
1	1.452(28)	7.240(70)	32.83(28)	4.672(64)	3.321(53)	3.287(54)	3.318(54)
2	1.449(28)	6.670(78)	32.33(28)	3.989(60)	3.003(54)	2.895(54)	2.943(54)
3	1.449(28)	6.093(79)	32.19(29)	3.498(54)	2.664(51)	2.519(49)	2.663(51)
5	1.449(28)	5.184(71)	31.83(29)	3.206(50)	2.299(47)	2.166(45)	2.289(47)
10	1.449(28)	4.603(64)	31.59(31)	3.029(47)	2.048(43)	2.008(41)	2.050(43)
20	1.449(28)	4.292(60)	30.97(32)	2.914(45)	1.954(39)	1.933(38)	1.961(40)
35	1.448(28)	4.157(57)	30.74(33)	2.850(44)	1.919(38)	1.897(37)	1.921(38)
50	1.448(28)	4.098(57)	30.54(34)	2.796(43)	1.887(37)	1.873(36)	1.902(38)
75	1.446(28)	4.017(55)	30.37(35)	2.757(42)	1.864(36)	1.855(35)	1.880(37)
100	1.446(28)	3.969(54)	30.24(35)	2.733(42)	1.852(35)	1.838(35)	1.871(36)

TABLE XI: SNR_{95} for the latent-only architecture (without reconstruction branch).

FP/yr	10^{-12}	10^{-11}	10^{-10}	10^{-9}	10^{-8}	10^{-7}	10^{-6}
0.1	2.953(34)	4.422(42)	19.56(32)	3.294(38)	2.177(28)	2.157(26)	2.171(28)
0.5	2.867(31)	4.331(42)	18.71(39)	3.201(38)	2.117(29)	2.115(29)	2.126(30)
1	2.849(31)	4.307(43)	18.26(45)	3.171(38)	2.094(31)	2.097(31)	2.095(30)
2	2.817(31)	4.273(43)	17.89(51)	3.150(38)	2.066(32)	2.067(32)	2.073(31)
3	2.786(30)	4.245(43)	17.83(48)	3.138(38)	2.055(33)	2.048(33)	2.054(32)
5	2.716(27)	4.205(43)	17.63(51)	3.101(37)	2.032(35)	2.037(34)	2.048(33)
10	2.661(27)	4.165(42)	17.54(45)	3.073(37)	2.001(35)	2.014(34)	2.038(33)
20	2.638(29)	4.136(43)	17.47(38)	3.034(37)	1.976(36)	1.993(35)	2.019(34)
35	2.625(30)	4.112(43)	17.23(35)	3.016(37)	1.949(37)	1.983(35)	2.003(35)
50	2.615(31)	4.082(43)	17.07(33)	2.993(37)	1.934(37)	1.971(35)	1.990(35)
75	2.603(31)	4.034(42)	17.01(31)	2.981(37)	1.916(36)	1.968(35)	1.985(35)
100	2.591(32)	4.018(42)	16.99(29)	2.969(37)	1.908(36)	1.958(35)	1.976(35)

Appendix D: Parameter-estimation prototype diagnostics

Figures 7 and 8 show the two diagnostics used to assess the prototype of Sec. VIII.

Figure 7 plots, for each estimated parameter, the predicted posterior mean μ against the true value; the error bars are the $\pm 1\sigma$ predictive width returned by the Gaussian head, and the red dashed line is the identity $\mu = y_{\text{true}}$ (perfect recovery). Each point is coloured by a per-window signal-strength proxy (the normalised clean-signal SNR): this colour bar is what makes the calibration visible at a glance, since it separates strong-signal windows (where a lot of a signal is present with the highest amplitude part), which cluster tightly on the identity line with small σ , from weak-signal windows (where the peak amplitude might not be inside), which scatter and carry correspondingly inflated σ . The behaviour is parameter-dependent: amplitude, SNR proxy and time-of-arrival track the identity well for strong signals, whereas the heaviest mass bin ($10^{-8} M_{\odot}$) is systematically compressed. This identifiability issue is further discussed in Sec. VIII.

Figure 8 is a physical-consistency check. The chirp rate and the PBH mass are not independent but linked by Eq. (13) ($k \propto m_{\text{PBH}}^{5/3}$), so the two heads, although trained separately, must reproduce a single locus in the $(\log_{10} m_{\text{PBH}}, \log_{10} k)$ plane. The predicted pairs (red) follow the same relation as the truth (black), confirming that the estimator is internally consistent rather than fitting the two quantities independently.

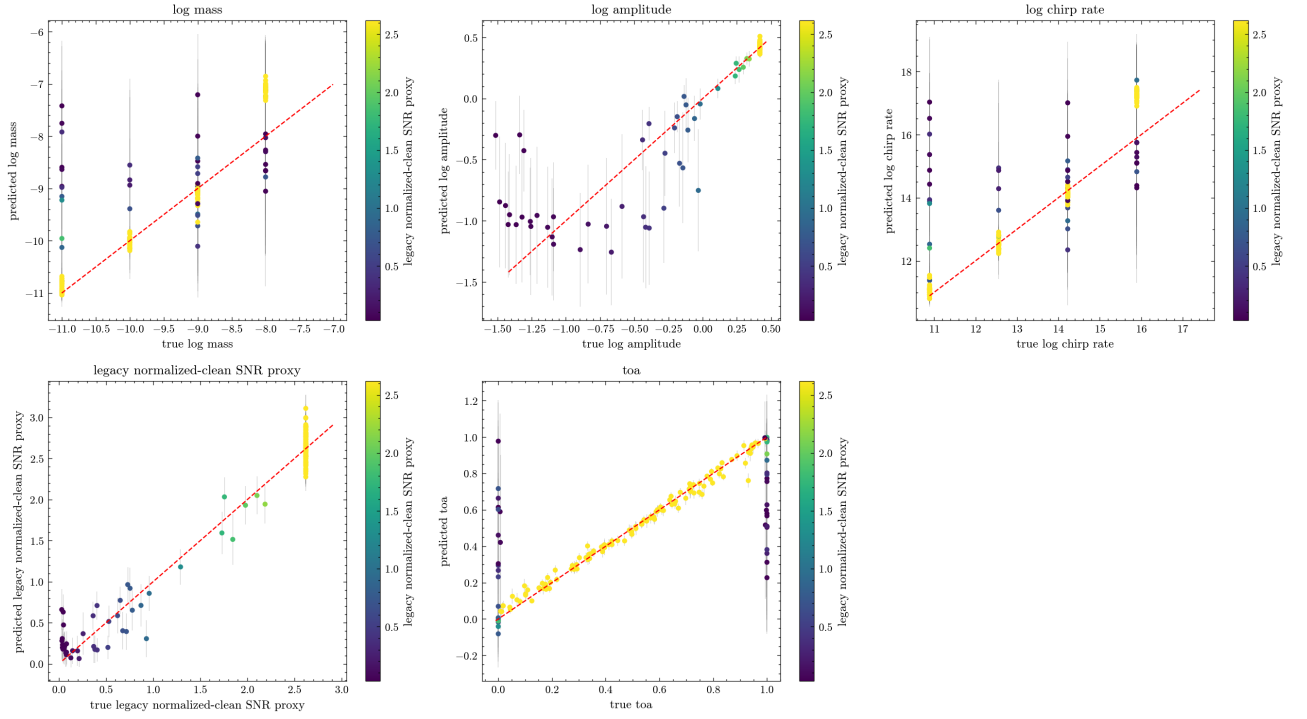


FIG. 7: Predicted vs. true parameters. Markers are posterior means μ , error bars the $\pm 1\sigma$ predictive width, and the red dashed line the identity. Colour encodes the per-window normalised clean-signal SNR proxy, showing that accuracy improves and σ shrinks with increasing signal strength.

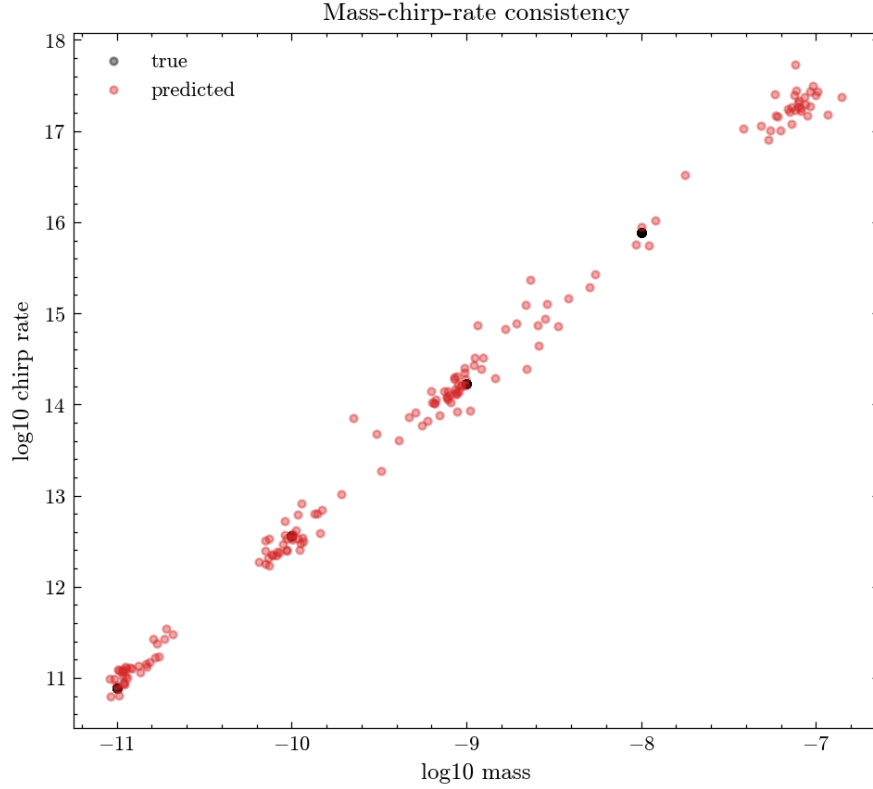


FIG. 8: Mass–chirp–rate consistency. True (black) and predicted (red) $(\log_{10} m_{\text{PBH}}, \log_{10} k)$ pairs. The predictions follow the physical relation of Eq. (13), confirming the two separately trained heads are mutually consistent.